

BLUE CITY

Analysis & Asset Selection

From no financial-analysis knowledge to a defensible investment decision

START

Understand the business

BUILD

Calculate and interpret

CAPABILITY

Compare, value and allocate

How Blue City teaches you

Every concept follows the same simple learning loop.

1

Understand

What does the concept mean in ordinary language?

2

Formula

What relationship are we measuring?

3

Substitute

Where do the numbers go?

4

Calculate

Work through the arithmetic without skipping steps.

5

Interpret

What does the result tell an investor?

6

Decide

What should we investigate or do next?

The calculation rule

Do not memorize answers. Learn how to build them.

1 • DEFINE

Name every variable.

Ask: dollars, shares, percentage, years, or another unit?

Know what the numerator and denominator represent.

2 • CALCULATE

Write the formula first.

Substitute the numbers.

Keep units consistent.

Convert percentages to decimals when multiplying.

3 • INTERPRET

Translate the number into words.

Compare it with history, competitors or a threshold.

State what the result does NOT prove.

A correct calculation with a poor interpretation is still weak investment analysis.

Your Blue City roadmap

Nine portals build one capability: selecting and valuing investments.



The same \$14,800 Portfolio Lab continues. Blue City asks: What deserves capital, at what price, and why?

Understanding Companies

Before ratios and valuation, understand how the business makes money.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

A company is an economic system

Start with the business — not the stock price.

WHAT IS SOLD?

Product, service, subscription, license, financing or marketplace access.

WHO PAYS?

Consumer, business, government, advertiser or subscriber.

WHAT DRIVES VOLUME?

Units, users, stores, contracts, occupancy or transactions.

WHAT DRIVES PRICE?

List price, mix, premium features, negotiated rates.

WHAT ARE THE COSTS?

Inputs, labor, technology, sales, rent, financing and capex.

WHY CAN IT WIN?

Cost, switching costs, network effects, brand, IP, data or distribution.

Revenue = price × quantity

A company sells 240,000 units at an average price of \$18.

FORMULA

$$\text{Revenue} = \text{Price} \times \text{Quantity}$$

WHAT THE FORMULA IS ASKING

This measures the total sales generated by price and volume.

WORK IT STEP BY STEP

Identify units: \$18 per unit and 240,000 units.

$$\text{Revenue} = \$18 \times 240,000$$

$$\text{Revenue} = \$4,320,000$$

ANSWER

\$4.32 million

WHAT IT MEANS

It tells us scale, not profit. Next ask what the units cost to produce and whether price or volume is driving growth.

Revenue growth

Revenue rises from \$48m to \$54m.

FORMULA

$$\text{Growth} = (\text{New revenue} - \text{Old revenue}) / \text{Old revenue} \times 100\%$$

WHAT THE FORMULA IS ASKING

The old revenue is the base, so it belongs in the denominator.

WORK IT STEP BY STEP

Change = \$54m - \$48m = \$6m
Relative change = \$6m / \$48m = 0.125
Percentage = 0.125 × 100% = 12.5%

ANSWER

12.5% growth

WHAT IT MEANS

Growth is meaningful only with context: margins, cash flow and market growth determine whether 12.5% is strong.

Market share

Company sales are \$900m in a \$6bn market.

FORMULA

$$\text{Market share} = \text{Company sales} / \text{Total market sales} \times 100\%$$

WHAT THE FORMULA IS ASKING

The numerator is the company; the denominator is the whole defined market.

WORK IT STEP BY STEP

Put both values in the same unit: \$0.9bn and \$6.0bn.

$$\$0.9\text{bn} / \$6.0\text{bn} = 0.15$$

$$0.15 \times 100\% = 15\%$$

ANSWER

15% market share

WHAT IT MEANS

A rising share can indicate competitive gains. It still does not tell us whether the company earns attractive profits.

Competitive advantage: can the economics persist?

Market share tells you where the company is. Advantage asks whether it can stay there.

COST

Lower unit cost

Evidence: Look for durable margins / scale

SWITCHING

Hard to leave

Evidence: Retention / low churn

NETWORK

More users → more value

Evidence: User growth and engagement

BRAND

Pricing power

Evidence: Price increases without major volume loss

IP

Protected know-how

Evidence: Patents / proprietary technology

DATA / DISTRIBUTION

Unique access

Evidence: Hard-to-replicate channels

Financial Statements

Translate the business model into revenue, profit, assets, liabilities and cash.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

The three statements answer different questions

Think of them as three views of the same company.

INCOME STATEMENT

Over a period:

How much did we sell?
What did it cost?
What accounting profit remained?

Revenue → Gross profit → Operating income →
Net income

BALANCE SHEET

At a point in time:

What do we own?
What do we owe?
What belongs to shareholders?

Assets = Liabilities + Equity

CASH-FLOW STATEMENT

Over a period:

Where did cash come from?
Where did cash go?

Operating + Investing + Financing cash flows

Build an income statement

Revenue \$12m; COGS \$7.2m; operating expenses \$2.4m; interest \$0.3m; tax \$0.42m.

FORMULA

Profit is calculated layer by layer

WHAT THE FORMULA IS ASKING

Each subtraction answers a different question about where revenue is consumed.

WORK IT STEP BY STEP

Gross profit = $\$12.0\text{m} - \$7.2\text{m} = \$4.8\text{m}$

Operating income = $\$4.8\text{m} - \$2.4\text{m} = \$2.4\text{m}$

Pre-tax income = $\$2.4\text{m} - \$0.3\text{m} = \$2.1\text{m}$

Net income = $\$2.1\text{m} - \$0.42\text{m} = \$1.68\text{m}$

ANSWER

\$1.68m net income

WHAT IT MEANS

Profitability exists, but the next question is whether the company actually generated cash.

The accounting equation

Assets are \$18m and liabilities are \$11m.

FORMULA

$$\text{Equity} = \text{Assets} - \text{Liabilities}$$

WHAT THE FORMULA IS ASKING

Equity is the accounting residual after liabilities are deducted from assets.

WORK IT STEP BY STEP

Equity = \$18m – \$11m
Equity = \$7m

ANSWER

\$7m shareholders' equity

WHAT IT MEANS

Book equity is not automatically the market value of the shares.

Working capital

Current assets are \$5.4m and current liabilities are \$3.9m.

FORMULA

Net working capital = Current assets – Current liabilities

WHAT THE FORMULA IS ASKING

Working capital focuses on short-term operating resources and obligations.

WORK IT STEP BY STEP

$\text{NWC} = \$5.4\text{m} - \3.9m
 $\text{NWC} = \$1.5\text{m}$

ANSWER

\$1.5m NWC

WHAT IT MEANS

Positive NWC can help operations, but inventory or receivables may still be slow or poor quality.

Free cash flow

Cash flow from operations is \$3.6m and capex is \$1.4m.

FORMULA

$$\text{FCF} = \text{Cash flow from operations} - \text{Capital expenditures}$$

WHAT THE FORMULA IS ASKING

The formula asks what operating cash remains after capital investment.

WORK IT STEP BY STEP

$$\begin{aligned}\text{FCF} &= \$3.6\text{m} - \$1.4\text{m} \\ \text{FCF} &= \$2.2\text{m}\end{aligned}$$

ANSWER

\$2.2m FCF

WHAT IT MEANS

FCF can fund debt repayment, dividends, buybacks, acquisitions or cash accumulation.

Why profit and cash can differ

Accrual accounting records economic activity before or after cash moves.

RECEIVABLES ↑

A sale can increase revenue and profit before the customer pays cash.

If receivables rise rapidly, cash may lag earnings.

INVENTORY ↑

Cash may be spent on inventory before it becomes an expense through a sale.

Inventory can consume cash.

CAPEX

Buying long-lived assets uses cash now, while accounting expense may appear gradually through depreciation.

Strong analysis asks: does cash flow confirm the accounting story?

Financial Health

Turn the statements into ratios that diagnose profitability, liquidity, leverage and capital efficiency.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Margins: follow one dollar of revenue

Each margin asks how much of sales survives a different layer of cost.

Revenue

\$1.00

Starting sales dollar

Gross margin

\$0.40

After direct costs

Operating margin

\$0.15

After operating costs

Net margin

\$0.10

After interest and tax

Profitability margins

Revenue \$20m; gross profit \$8m; operating income \$3m; net income \$2m.

FORMULA

$$\text{Margin} = \text{Profit measure} / \text{Revenue} \times 100\%$$

WHAT THE FORMULA IS ASKING

The same revenue denominator lets us see how much survives each cost layer.

WORK IT STEP BY STEP

Gross = $\$8\text{m} / \$20\text{m} \times 100\% = 40\%$

Operating = $\$3\text{m} / \$20\text{m} \times 100\% = 15\%$

Net = $\$2\text{m} / \$20\text{m} \times 100\% = 10\%$

ANSWER

40% • 15% • 10%

WHAT IT MEANS

Compare margins over time and with similar businesses. A margin is not 'good' without a relevant benchmark.

Current ratio

Current assets \$6m; current liabilities \$4m.

FORMULA

$$\text{Current ratio} = \text{Current assets} / \text{Current liabilities}$$

WHAT THE FORMULA IS ASKING

This compares short-term accounting resources with short-term obligations.

WORK IT STEP BY STEP

Current ratio = \$6m / \$4m
Current ratio = 1.50×

ANSWER

1.50×

WHAT IT MEANS

It suggests \$1.50 of current assets per \$1 of current liabilities, but asset quality matters.

Quick ratio

Cash \$1m + securities \$0.4m + receivables \$1.8m; current liabilities \$4m.

FORMULA

$$\text{Quick ratio} = \frac{(\text{Cash} + \text{Securities} + \text{Receivables})}{\text{Current liabilities}}$$

WHAT THE FORMULA IS ASKING

Inventory is removed to focus on more liquid current assets.

WORK IT STEP BY STEP

Quick assets = \$1.0m + \$0.4m + \$1.8m = \$3.2m

Quick ratio = \$3.2m / \$4.0m

Quick ratio = 0.80×

ANSWER

0.80×

WHAT IT MEANS

A company can have an acceptable current ratio but a weak quick ratio if inventory dominates current assets.

Debt-to-equity and interest coverage

Debt \$12m, equity \$8m; EBIT \$4.5m, interest \$1m.

FORMULA

$$D/E = \text{Debt} / \text{Equity} \quad \bullet \quad \text{Coverage} = \text{EBIT} / \text{Interest}$$

WHAT THE FORMULA IS ASKING

D/E measures financing structure; coverage measures ability to service interest from operating profit.

WORK IT STEP BY STEP

$$\begin{aligned} D/E &= \$12\text{m} / \$8\text{m} = 1.50\times \\ \text{Coverage} &= \$4.5\text{m} / \$1.0\text{m} = 4.5\times \end{aligned}$$

ANSWER

$$1.50\times D/E \bullet 4.5\times \text{coverage}$$

WHAT IT MEANS

Debt can amplify shareholder returns and shareholder risk. Use both ratios together.

Return on equity

Net income \$2.4m; beginning equity \$10m; ending equity \$14m.

FORMULA

$$\text{ROE} = \text{Net income} / \text{Average equity} \times 100\%$$

WHAT THE FORMULA IS ASKING

Average equity matches a period's income with the capital employed across that period.

WORK IT STEP BY STEP

Average equity = $(\$10\text{m} + \$14\text{m}) / 2 = \$12\text{m}$

ROE = $\$2.4\text{m} / \$12\text{m} \times 100\%$

ROE = 20%

ANSWER

20% ROE

WHAT IT MEANS

High ROE can reflect strong economics — or a small equity base caused by heavy leverage.

Return on invested capital

EBIT \$5m; tax rate 25%; average invested capital \$20m.

FORMULA

$$\text{NOPAT} = \text{EBIT}(1 - \text{tax}) \quad \bullet \quad \text{ROIC} = \text{NOPAT} / \text{Invested capital} \times 100\%$$

WHAT THE FORMULA IS ASKING

ROIC asks how efficiently the operating business uses invested capital after tax.

WORK IT STEP BY STEP

$\text{NOPAT} = \$5\text{m} \times (1 - 0.25) = \3.75m
 $\text{ROIC} = \$3.75\text{m} / \$20\text{m} \times 100\%$
 $\text{ROIC} = 18.75\%$

ANSWER

18.75% ROIC

WHAT IT MEANS

It is especially useful when comparing economic quality and reinvestment, but definitions of invested capital must be consistent.

Financial-health red flags

Ratios are strongest when several signals tell the same story.

RED FLAG 1

Receivables grow much faster than revenue

RED FLAG 2

Profit rises while operating cash flow falls

RED FLAG 3

Debt rises while coverage weakens

RED FLAG 4

Margins deteriorate without explanation

RED FLAG 5

Inventory grows much faster than sales

RED FLAG 6

Share count rises rapidly

RED FLAG 7

Adjusted profit diverges from statutory profit

RED FLAG 8

High ROE is mainly leverage-driven

Company Valuation

Move from 'Is this a good company?' to 'What might it be worth, and what price is defensible?'

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Price is observable. Value is estimated.

Valuation is a disciplined argument built from assumptions.

PRICE

What the market currently asks.

Observable now.

Can move quickly with expectations and sentiment.

VALUE

An estimate based on earnings, cash flows, assets, growth and risk.

Depends on assumptions.

MARGIN OF SAFETY

A gap between estimated value and purchase price designed to allow for uncertainty.

Not a guarantee.

P/E ratio

Share price \$48; EPS \$3.20.

FORMULA

$$\text{P/E} = \text{Share price} / \text{Earnings per share}$$

WHAT THE FORMULA IS ASKING

P/E tells how many dollars of price investors pay for each dollar of current EPS.

WORK IT STEP BY STEP

$$\text{P/E} = \$48 / \$3.20$$

$$\text{P/E} = 15.0\times$$

ANSWER

15.0× earnings

WHAT IT MEANS

A lower P/E is not automatically cheaper: growth, quality, leverage and cyclicalities can justify differences.

PEG ratio

P/E is 24x; expected EPS growth is 12%.

FORMULA

$$\text{PEG} = \text{P/E} / \text{Expected EPS growth rate (\%)}$$

WHAT THE FORMULA IS ASKING

PEG relates a valuation multiple to an expected growth rate using the convention in the notes.

WORK IT STEP BY STEP

$$\begin{aligned}\text{PEG} &= 24 / 12 \\ \text{PEG} &= 2.0\end{aligned}$$

ANSWER

$$\text{PEG} = 2.0$$

WHAT IT MEANS

It is a shortcut. Forecast growth can be wrong and two companies with the same PEG can have very different quality.

Price-to-book

Equity \$500m; 100m shares; share price \$7.50.

FORMULA

$$\text{BVPS} = \text{Equity} / \text{Shares} \quad \bullet \quad \text{P/B} = \text{Price} / \text{BVPS}$$

WHAT THE FORMULA IS ASKING

First calculate book value per share so price and book value use the same per-share unit.

WORK IT STEP BY STEP

$$\text{BVPS} = \$500\text{m} / 100\text{m} = \$5.00$$

$$\text{P/B} = \$7.50 / \$5.00$$

$$\text{P/B} = 1.50\times$$

ANSWER

1.50× book value

WHAT IT MEANS

P/B can be useful for asset-heavy or financial firms, but less revealing for intangible-heavy businesses.

Enterprise value and EV/EBITDA

Market cap \$900m; debt \$300m; cash \$100m; EBITDA \$125m.

FORMULA

$$\text{EV} = \text{Market cap} + \text{Debt} - \text{Cash} \quad \bullet \quad \text{EV/EBITDA} \\ = \text{EV} / \text{EBITDA}$$

WHAT THE FORMULA IS ASKING

Enterprise value looks at the operating business across debt and equity financing.

WORK IT STEP BY STEP

$$\begin{aligned} \text{EV} &= \$900\text{m} + \$300\text{m} - \$100\text{m} = \$1.10\text{bn} \\ \text{EV/EBITDA} &= \$1.10\text{bn} / \$125\text{m} \\ &= \$1,100\text{m} / \$125\text{m} = 8.8\times \end{aligned}$$

ANSWER

8.8× EBITDA

WHAT IT MEANS

This multiple can improve capital-structure comparability, but EBITDA is not free cash flow.

Free-cash-flow yield

FCF \$72m; market cap \$900m.

FORMULA

$$\text{FCF yield} = \text{FCF} / \text{Market capitalization} \times 100\%$$

WHAT THE FORMULA IS ASKING

This expresses current free cash generation relative to the equity market price.

WORK IT STEP BY STEP

$$\begin{aligned}\text{FCF yield} &= \$72\text{m} / \$900\text{m} \times 100\% \\ &= 0.08 \times 100\% \\ &= 8.0\%\end{aligned}$$

ANSWER

8.0% FCF yield

WHAT IT MEANS

A high yield can be attractive or can signal that investors expect cash flow to decline. Sustainability matters.

DCF: value future cash in today's dollars

DCF is conceptually simple: forecast cash, then discount it for time and risk.

1

Forecast FCF

Estimate cash flows for explicit years.

2

Choose discount rate

Higher required return → lower present value.

3

Discount each year

$CF_t / (1+r)^t$

4

Estimate terminal value

Value beyond explicit forecast.

5

Discount terminal value

Bring it back to today.

6

Sum the pieces

Estimate enterprise value.

FORMULA

$$PV(CF_t) = CF_t / (1+r)^t$$

FORMULA

$$TV = FCF_{n+1} / (r-g)$$

WHY ASSUMPTIONS MATTER

Small changes in r or g can create large changes in estimated value.

Three-year DCF

FCF: \$10m, \$12m, \$14m. Discount rate 9%. Ignore terminal value.

FORMULA

$$PV = \sum CF_t / (1+r)^t$$

WHAT THE FORMULA IS ASKING

Each year's cash flow is discounted for a different number of years.

WORK IT STEP BY STEP

Year 1: $\$10\text{m} / (1.09) = \9.174m
Year 2: $\$12\text{m} / (1.09)^2 = \10.100m
Year 3: $\$14\text{m} / (1.09)^3 = \10.811m
Total = \$30.085m

ANSWER

\$30.085m present value

WHAT IT MEANS

This is only the three explicit cash flows. A going concern normally needs a terminal-value assumption.

Terminal value

Year-5 FCF \$20m; $g=3\%$; $r=9\%$.

FORMULA

$$FCF_6 = FCF_5(1+g) \quad \bullet \quad TV_5 = FCF_6/(r-g)$$

WHAT THE FORMULA IS ASKING

The Gordon-growth formula assumes cash flow grows perpetually at g after the forecast period.

WORK IT STEP BY STEP

$$\begin{aligned} FCF_6 &= \$20m \times 1.03 = \$20.6m \\ r-g &= 0.09 - 0.03 = 0.06 \\ TV_5 &= \$20.6m / 0.06 = \$343.33m \end{aligned}$$

ANSWER

\$343.33m at end of Year 5

WHAT IT MEANS

Because terminal value can dominate a DCF, r and g deserve careful sensitivity testing.

Margin of safety

Estimated value \$60; market price \$45.

FORMULA

$$\text{MOS} = (\text{Estimated value} - \text{Market price}) / \text{Estimated value} \times 100\%$$

WHAT THE FORMULA IS ASKING

The denominator is estimated value because the discount is measured relative to what you think the asset is worth.

WORK IT STEP BY STEP

Difference = \$60 - \$45 = \$15
\$15 / \$60 = 0.25
0.25 × 100% = 25%

ANSWER

25% margin of safety

WHAT IT MEANS

The margin compensates for uncertainty only if the value estimate itself is reasonable.

Stock Selection

Combine business quality, financial health and valuation into a stock-selection process.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Seven stock-selection lenses

Different styles ask different questions.

GROWTH

Can earnings compound rapidly?

VALUE

Is price low vs normalized economics?

QUALITY

Is the business durable and efficient?

DIVIDEND

Is income sustainable?

MOMENTUM

Are trends persistent?

CONTRARIAN

Is consensus too pessimistic?

GARP

Is growth reasonably priced?

Dividend yield and payout

Dividend \$1.80/share; price \$45. Total dividends \$24m; net income \$60m.

FORMULA

$$\text{Yield} = \text{DPS} / \text{Price} \times 100\% \quad \bullet \quad \text{Payout} = \frac{\text{Dividends}}{\text{Net income}} \times 100\%$$

WHAT THE FORMULA IS ASKING

Yield relates cash income per share to the price paid; payout relates distributions to accounting profit.

WORK IT STEP BY STEP

Yield = $\$1.80 / \$45 \times 100\% = 4.0\%$
Payout = $\$24\text{m} / \$60\text{m} \times 100\% = 40\%$

ANSWER

4.0% yield • 40% payout

WHAT IT MEANS

A high yield is not automatically safe. Check cash flow, debt and whether the dividend is sustainable.

Position sizing

Portfolio \$14,800; target weight 8%; share price \$37.

FORMULA

$$\text{Position value} = \text{Portfolio} \times \text{Weight} \quad \bullet \quad \text{Shares} = \frac{\text{Position value}}{\text{Price}}$$

WHAT THE FORMULA IS ASKING

Sizing converts a portfolio percentage into an actual dollar and share amount.

WORK IT STEP BY STEP

Position = $\$14,800 \times 0.08 = \$1,184$
Shares = $\$1,184 / \$37 = 32$

ANSWER

$\$1,184 \bullet 32$ shares

WHAT IT MEANS

A strong idea can still be a poor portfolio decision if the position is too large.

A beginner's stock-selection workflow

Do the work in this order to avoid starting with the share price.

1

Business

How does it make money?

2

Financial health

Margins, liquidity, leverage, ROE/ROIC, FCF.

3

Competitive position

Why might economics persist?

4

Valuation

What are you paying? Use suitable methods.

5

Catalyst

What could change expectations?

6

Risk

What could falsify the thesis?

7

Position size

How much portfolio risk should this idea receive?

Buying an Entire Company

Move from buying shares to evaluating control of a private operating business.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Equity value is not enterprise value

An acquirer must understand the whole capital structure.

FORMULA

$$\text{Enterprise value} = \text{Equity value} + \text{Debt} - \text{Cash}$$

EQUITY VALUE

What belongs to the owners.

DEBT

Financing claims that affect the total business value.

CASH

Reduces net purchase burden in the simplified EV bridge.

Enterprise value in an acquisition

Seller wants \$4.0m equity; debt \$1.2m; cash \$0.3m.

FORMULA

$$\text{EV} = \text{Equity value} + \text{Debt} - \text{Cash}$$

WHAT THE FORMULA IS ASKING

The formula bridges what equity owners receive to the value of the operating business across financing.

WORK IT STEP BY STEP

$$\begin{aligned}\text{EV} &= \$4.0\text{m} + \$1.2\text{m} - \$0.3\text{m} \\ \text{EV} &= \$4.9\text{m}\end{aligned}$$

ANSWER

\$4.9m enterprise value

WHAT IT MEANS

The business effectively costs more than the equity cheque because debt is part of the capital structure.

Normalize private-company earnings

Reported EBITDA \$700k; valid legal add-back \$80k; owner salary is \$60k below market.

FORMULA

Normalized EBITDA = Reported EBITDA + Valid add-backs – Missing recurring costs

WHAT THE FORMULA IS ASKING

Normalization attempts to estimate sustainable economics rather than blindly capitalizing reported profit.

WORK IT STEP BY STEP

$\$700k + \$80k - \$60k$
 $= \$720k$

ANSWER

\$720,000 normalized EBITDA

WHAT IT MEANS

Add-backs require evidence. Aggressive normalization can manufacture an unrealistically high valuation.

Acquisition valuation multiple

Normalized EBITDA \$720k; selected multiple 5.5×

FORMULA

$$\text{EV} = \text{Normalized EBITDA} \times \text{Selected multiple}$$

WHAT THE FORMULA IS ASKING

The multiple converts a sustainable earnings measure into a business value estimate.

WORK IT STEP BY STEP

$$\begin{aligned}\text{EV} &= \$720,000 \times 5.5 \\ \text{EV} &= \$3,960,000\end{aligned}$$

ANSWER

\$3.96m enterprise value

WHAT IT MEANS

The selected multiple must reflect growth, customer concentration, cyclicality, quality and comparable transactions.

Acquisition financing

Purchase price \$4.0m; lender finances 60%.

FORMULA

$$\text{Debt} = \text{Price} \times \text{Debt \%} \bullet \text{Equity} = \text{Price} - \text{Debt}$$

WHAT THE FORMULA IS ASKING

Financing determines how much buyer capital is required upfront.

WORK IT STEP BY STEP

$$\begin{aligned}\text{Debt} &= \$4.0\text{m} \times 0.60 = \$2.4\text{m} \\ \text{Equity} &= \$4.0\text{m} - \$2.4\text{m} = \$1.6\text{m}\end{aligned}$$

ANSWER

\$2.4m debt • \$1.6m equity

WHAT IT MEANS

Debt can improve equity returns if the business performs, but creates fixed repayment risk.

Debt-service coverage

Cash available for debt service \$600k; required debt service \$400k.

FORMULA

$$\text{DSCR} = \frac{\text{Cash available for debt service}}{\text{Required debt service}}$$

WHAT THE FORMULA IS ASKING

DSCR measures the cushion between cash available and required debt payments.

WORK IT STEP BY STEP

$$\begin{aligned}\text{DSCR} &= \$600,000 / \$400,000 \\ \text{DSCR} &= 1.50\times\end{aligned}$$

ANSWER

1.50× DSCR

WHAT IT MEANS

A lower ratio leaves less room for an earnings downturn or unexpected cash need.

Due diligence: valuation is not enough

A cheap business can be expensive if hidden risks destroy the economics.

REVENUE

Recurring? Customer concentration?

EARNINGS

Sustainable? One-offs?

WORKING CAPITAL

How much cash is needed?

LEGAL / TAX

Liens, lawsuits, liabilities?

OPERATIONS

Key staff, suppliers, systems?

SELLER DEPENDENCE

Can it operate without the owner?

ASSETS

Condition and maintenance backlog?

DEAL TERMS

Earn-out, warranties, indemnities?

Real Estate Investing

Underwrite property income, valuation and financing instead of relying on the listing price.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Property cash flow: from rent to NOI

Start with potential rent, then remove realistic operating losses and costs.

1

Gross potential rent

What would rent be at full occupancy?

2

Vacancy / credit loss

Not every unit is occupied and every rent collected.

3

Other income

Parking, laundry or other property income.

4

Effective gross income

Income after vacancy plus other income.

5

Operating expenses

Property operating costs.

6

NOI

Income before financing and tax.

FORMULA

$$\text{NOI} = \text{Effective gross income} - \text{Operating expenses}$$

IMPORTANT

NOI is before mortgage payments. Financing is analyzed separately.

Gross rent and vacancy

Four units at \$1,500/month; expected vacancy 5%.

FORMULA

$$\text{GPR} = \text{Units} \times \text{Monthly rent} \times 12$$
$$\text{Vacancy loss} = \text{GPR} \times \text{Vacancy rate}$$

WHAT THE FORMULA IS ASKING

Potential rent assumes full occupancy; vacancy makes the estimate more realistic.

WORK IT STEP BY STEP

$$\text{GPR} = 4 \times \$1,500 \times 12 = \$72,000$$

$$\text{Vacancy loss} = \$72,000 \times 0.05 = \$3,600$$

$$\text{Effective rental income} = \$72,000 - \$3,600 = \$68,400$$

ANSWER

\$68,400 after vacancy

WHAT IT MEANS

Vacancy assumptions should reflect the property and local market, not an optimistic guess.

NOI

Effective gross income \$70,000; operating expenses \$26,000.

FORMULA

$$\text{NOI} = \text{Effective gross income} - \text{Operating expenses}$$

WHAT THE FORMULA IS ASKING

NOI isolates property operating performance before financing.

WORK IT STEP BY STEP

$$\begin{aligned}\text{NOI} &= \$70,000 - \$26,000 \\ \text{NOI} &= \$44,000\end{aligned}$$

ANSWER

\$44,000 NOI

WHAT IT MEANS

It becomes a central input for cap-rate valuation.

Cap rate

NOI \$44,000; property price \$550,000.

FORMULA

$$\text{Cap rate} = \text{NOI} / \text{Property value} \times 100\%$$

WHAT THE FORMULA IS ASKING

Cap rate expresses current NOI as a percentage of property value.

WORK IT STEP BY STEP

$$\begin{aligned}\text{Cap rate} &= \$44,000 / \$550,000 \times 100\% \\ &= 0.08 \times 100\% \\ &= 8.0\%\end{aligned}$$

ANSWER

8.0% cap rate

WHAT IT MEANS

Compare only with relevant properties and consider location, quality, growth and risk.

Value from a cap rate

NOI \$60,000; market cap rate 7.5%.

FORMULA

$$\text{Property value} = \text{NOI} / \text{Cap rate}$$

WHAT THE FORMULA IS ASKING

Rearranging the cap-rate formula converts income and a required market yield into a value estimate.

WORK IT STEP BY STEP

Convert 7.5% to decimal: 0.075
Value = \$60,000/0.075
Value = \$800,000

ANSWER

\$800,000 estimated value

WHAT IT MEANS

For the same NOI, a lower cap rate implies a higher value.

Loan-to-value

Property price \$600,000; LTV 70%.

FORMULA

$$\text{Loan} = \text{Price} \times \text{LTV} \quad \bullet \quad \text{Equity} = \text{Price} - \text{Loan}$$

WHAT THE FORMULA IS ASKING

LTV shows how much of the property value is financed with debt.

WORK IT STEP BY STEP

$$\begin{aligned}\text{Loan} &= \$600,000 \times 0.70 = \$420,000 \\ \text{Equity} &= \$600,000 - \$420,000 = \$180,000\end{aligned}$$

ANSWER

\$420k loan • \$180k equity

WHAT IT MEANS

Leverage reduces initial equity but magnifies both gains and losses on that equity.

Cash-on-cash return

NOI \$54,000; debt service \$30,000; initial cash \$180,000.

FORMULA

$$\text{CoC} = \text{Annual pre-tax cash flow} / \text{Cash invested} \times 100\%$$

WHAT THE FORMULA IS ASKING

This focuses on annual cash received relative to actual cash equity invested.

WORK IT STEP BY STEP

Cash flow = \$54,000 - \$30,000 = \$24,000
CoC = \$24,000 / \$180,000 × 100%
= 13.33%

ANSWER

13.33% cash-on-cash return

WHAT IT MEANS

It is not total return: appreciation and principal repayment are separate.

Forex Investing

Learn quote conventions, currency movement, pips, spreads and leverage before taking a simulated FX position.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Read the currency pair first

If EUR/USD = 1.1000, one euro costs 1.10 U.S. dollars.

EUR

BASE CURRENCY

The unit being priced.

1 EUR is the reference unit.

/

USD

QUOTE CURRENCY

The currency used to express the price.

1.1000 USD per EUR.

1.1000

THE RATE

How many USD buy 1 EUR.

Currency conversion

EUR/USD = 1.1000. Convert €8,000 to USD.

FORMULA

**Quote-currency value = Base-currency amount
× Exchange rate**

WHAT THE FORMULA IS ASKING

Because the quote is dollars per euro, multiply euros by dollars-per-euro.

WORK IT STEP BY STEP

USD = €8,000 × 1.1000
USD = \$8,800

ANSWER

\$8,800

WHAT IT MEANS

Always read the quote convention before choosing multiplication or division.

Percentage currency move

EUR/USD rises from 1.1000 to 1.1550.

FORMULA

$$\% \text{ change} = (\text{New rate} - \text{Old rate}) / \text{Old rate} \times 100\%$$

WHAT THE FORMULA IS ASKING

The old exchange rate is the starting base.

WORK IT STEP BY STEP

Change = $1.1550 - 1.1000 = 0.0550$
 $0.0550 / 1.1000 = 0.05$
 $0.05 \times 100\% = 5.0\%$

ANSWER

5.0% rise in EUR/USD

WHAT IT MEANS

Under this quote, a rise means the euro appreciated relative to the dollar.

Pips

GBP/USD moves from 1.2700 to 1.2765. Pip size = 0.0001.

FORMULA

$$\text{Pip movement} = (\text{New rate} - \text{Old rate}) / 0.0001$$

WHAT THE FORMULA IS ASKING

A pip standardizes small FX price movements.

WORK IT STEP BY STEP

Difference = $1.2765 - 1.2700 = 0.0065$
Pips = $0.0065 / 0.0001$
= 65

ANSWER

65 pips

WHAT IT MEANS

Pip conventions vary for some pairs, so confirm the pair's convention before calculating.

FX spread

EUR/USD bid 1.1048; ask 1.1051.

FORMULA

$$\text{Spread} = \text{Ask} - \text{Bid} \quad \bullet \quad \text{Spread pips} = \text{Spread} / 0.0001$$

WHAT THE FORMULA IS ASKING

The spread is the difference between the buying and selling quote.

WORK IT STEP BY STEP

Spread = 1.1051 - 1.1048 = 0.0003
Pips = 0.0003 / 0.0001
= 3

ANSWER

3-pip spread

WHAT IT MEANS

It is an immediate trading friction and matters more when trading frequently or targeting small moves.

Leverage and notional exposure

\$1,000 equity supports a 5× leveraged FX position.

FORMULA

Notional exposure = Account equity × Leverage

WHAT THE FORMULA IS ASKING

Notional is the market exposure controlled, not the cash equity committed.

WORK IT STEP BY STEP

Notional = \$1,000×5
Notional = \$5,000

ANSWER

\$5,000 notional exposure

WHAT IT MEANS

Leverage magnifies the impact of currency moves on the investor's equity.

Leveraged P/L

Notional exposure \$5,000; adverse currency move 2%.

FORMULA

$$\text{Approximate P/L} = \text{Notional exposure} \times \text{Percentage move}$$

WHAT THE FORMULA IS ASKING

The currency moved only 2%, but leverage makes the equity impact much larger.

WORK IT STEP BY STEP

$$\text{P/L} = \$5,000 \times (-0.02)$$

$$\text{P/L} = -\$100$$

$$\text{Loss on \$1,000 equity} = \$100 / \$1,000 = 10\%$$

ANSWER

$$-\$100 = -10\% \text{ of equity}$$

WHAT IT MEANS

This is why leverage changes risk, not just potential return.

What moves currencies?

FX is a relative macroeconomic market: one currency is always compared with another.

RATES

Interest-rate differentials

CENTRAL BANKS

Policy and guidance

INFLATION

Real return and policy

GROWTH

Relative economic strength

TRADE FLOWS

External demand/supply

RISK SENTIMENT

Safe-haven / risk appetite

POLITICS

Policy and geopolitical risk

EXPECTATIONS

What the market already priced

Blue City Investment Committee

Compare different assets using one decision framework, then allocate capital and defend the choice.

FROM CONCEPT → CALCULATION → INTERPRETATION → DECISION

Different assets create returns differently

Use a common decision language before comparing headline percentages.

PUBLIC STOCK

Return source:
Earnings • dividends • valuation

Liquidity:
Usually liquid

Main risk:
Business + valuation

PRIVATE BUSINESS

Return source:
Cash flow • growth • exit

Liquidity:
Low liquidity

Main risk:
Concentration + execution +
debt

REAL ESTATE

Return source:
Rent • NOI • appreciation

Liquidity:
Low direct liquidity

Main risk:
Vacancy + rates + leverage

FOREX

Return source:
Exchange-rate move • carry

Liquidity:
High in major pairs

Main risk:
Macro + leverage + policy

Expected return

Stock: 30% chance +30%; 50% chance +10%; 20% chance -20%.

FORMULA

$$E(R) = \Sigma [\text{Probability}_i \times \text{Return}_i]$$

WHAT THE FORMULA IS ASKING

Expected return is a probability-weighted planning estimate across scenarios.

WORK IT STEP BY STEP

$$0.30 \times 30\% = 9\%$$

$$0.50 \times 10\% = 5\%$$

$$0.20 \times (-20\%) = -4\%$$

$$\text{Total} = 9\% + 5\% - 4\% = 10\%$$

ANSWER

10% expected return

WHAT IT MEANS

It is not a promise. Scenario probabilities and returns are assumptions that should be challenged.

Opportunity cost

Stock expected return 10%; property expected return 6.5%.

FORMULA

**Incremental expected return = Chosen asset
 $E(R) - \text{Alternative } E(R)$**

WHAT THE FORMULA IS ASKING

Opportunity cost makes the rejected alternative explicit.

WORK IT STEP BY STEP

10.0% – 6.5%
= 3.5 percentage points

ANSWER

3.5 percentage points

WHAT IT MEANS

The higher expected return still may not be preferable if risk, liquidity or evidence quality is worse.

Allocate deployable capital

\$6,000 deployable: stock 35%, business 20%, property 30%, forex 15%.

FORMULA

Capital allocated = Deployable capital × Target allocation %

WHAT THE FORMULA IS ASKING

Weights convert strategy into actual dollars and must reconcile to available capital.

WORK IT STEP BY STEP

Stock = $\$6,000 \times 0.35 = \$2,100$

Business = $\$6,000 \times 0.20 = \$1,200$

Property = $\$6,000 \times 0.30 = \$1,800$

Forex = $\$6,000 \times 0.15 = \900

Check: $\$2,100 + \$1,200 + \$1,800 + \$900 = \$6,000$

ANSWER

\$6,000 fully allocated

WHAT IT MEANS

Allocation is a risk decision: check concentration, liquidity, leverage and overlap with existing Purple City holdings.

Investment Committee: defend the process

A strong answer is evidence-based, numerically clear and honest about uncertainty.

- 1 Opportunity**
How does this asset make money?
- 2 Evidence**
What are the strongest facts?
- 3 Financial health**
What do the key calculations say?
- 4 Valuation**
What are you paying and what assumptions matter?
- 5 Position size**
How much capital and why?
- 6 Downside**
What is the most important failure scenario?
- 7 Alternative**
Why this instead of the next-best opportunity?

Integrated practice: analyze before you decide

Use the same sequence across every Blue City opportunity.

1 • BUSINESS / ASSET

What creates the return?

What economic drivers matter?

What could permanently damage the thesis?

2 • NUMBERS

Calculate the relevant operating, financial-health and valuation measures.

Show formulas and substitutions.

3 • DECISION

Compare with alternatives.

Choose entry price and position size.

State what would change your mind.

Calculation self-check

Before accepting any answer, ask these questions.

CHECK 1

Did I write the correct formula?

CHECK 2

Are all units consistent?

CHECK 3

Did I convert percentages correctly?

CHECK 4

Did I use the correct starting-period denominator?

CHECK 5

Did I avoid rounding too early?

CHECK 6

Can I explain the answer in one sentence?

CHECK 7

What is the right benchmark?

CHECK 8

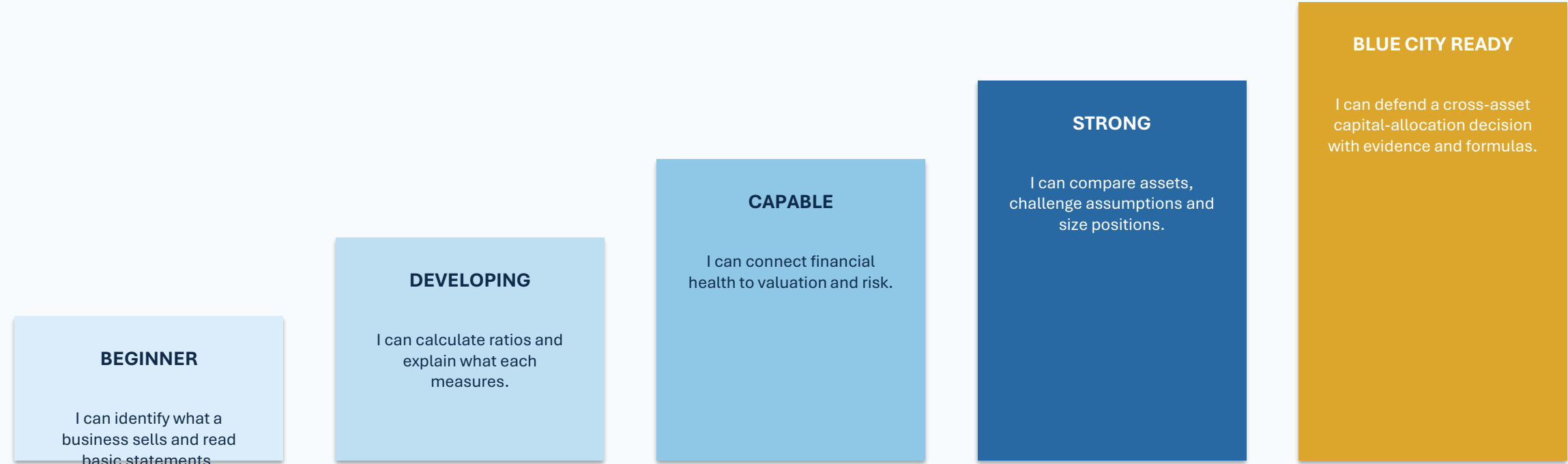
What important fact is still missing?

CHECK 9

Would a different assumption materially change the decision?

Blue City capability ladder

The goal is not expert jargon. It is progressively stronger judgment.



You are ready to analyze before you allocate.

The core skill is not finding a single 'best' number. It is connecting business economics, calculations, valuation, risk and portfolio fit.

1

Understand

2

Show the

3

Work the numbers

4

Interpret

5

Compare

6

Defend the decision

Next: Green City — Portfolio Management • manage the portfolio as conditions change.